Angelo Bitraj

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Summary

- First Class Honours graduate in Computer Science (Software Engineering) from Brunel University London, awarded the Brunel Software Innovation Award 2026 for best final year project.
- Experienced in back-end development with Java, Spring Boot, Python and FastAPI, building REST APIs, authentication flows and real-time WebSocket services.
- Skilled in front-end development with React and React Native, delivering cross-platform mobile and web interfaces tested with real users.
- Applied machine learning to a working product by fine-tuning a T5-small transformer, cutting Character Error Rate by 61.8% across a systematic eight-pipeline evaluation.
- Confident with MongoDB, PostgreSQL and Docker, handling datasets of 96,000+ records within a deployed real-time web application.
- Led a team of five to an A* grade, rotating as Scrum Master and Product Owner and running sprint planning in Jira using MoSCoW prioritisation.
- Developed leadership and communication skills across seven years in fine dining, supervising shifts of 25–30 staff and training over 100 team members.

Education

BSc Computer Science (Software Engineering) — First Class Honours

Brunel University London | September 2023 – July 2026

Key modules: Object-Oriented Programming (Java), Data Structures and Algorithms, Databases (SQL and NoSQL), Software Metrics, Usability Engineering, Agile Methodologies, Artificial Intelligence, Cybersecurity, Operating Systems, Distributed Systems, TCP/IP Networking.

Final year dissertation: SafeScan, an AI-powered mobile food safety application. The research contribution was a systematic evaluation of eight OCR pipelines, benchmarked on Character Error Rate, Word Error Rate and ingredient coverage.

Languages

English (Fluent), Italian (Fluent), Albanian (Native).

Project Work

SafeScan – AI Food Safety Mobile App (Final year dissertation, Grade A)

- Built a cross-platform mobile app with React Native, FastAPI, PostgreSQL and Docker that scans food labels, extracts ingredients via OCR and flags toxicological risks in real time.
- Designed and evaluated eight OCR pipelines as the primary research contribution, achieving a 61.8% reduction in Character Error Rate and 88.1% ingredient coverage against the baseline.
- Fine-tuned a T5-small transformer for post-OCR correction and integrated it into the FastAPI backend, improving extraction quality on low-resolution label images.
- Engineered a 50,000-pair synthetic training dataset and a chunked inference pipeline, enabling the model to run on low-memory mobile devices without sacrificing accuracy.

- Conducted UEQ usability testing with five participants, with all six scales scoring above the positive threshold.
- Awarded the Brunel Software Innovation Award 2026 by the Department of Computer Science, presented by Professor Kate Hone, Head of Department.

Crime Catchers – Real-Time Crime Reporting Web App (Team of five, Grade A*)

- Proposed the original project concept, led the Business Model Canvas analysis and aligned the product with UN SDG 16 targets 16.1.3 and 16.1.4.
- Built the Java and Spring Boot backend with RESTful APIs and MongoDB integration to handle 96,000+ crime records, implementing real-time proximity alerts using WebSockets.
- Developed the React.js front end for crime reporting with map pins, legal-notice safeguards and subscription-based email notifications via Google SMTP.
- Led sprint planning in Jira using MoSCoW prioritisation, rotating as Product Owner and Scrum Master across sprints to keep a team of five to deadline.
- Wrote JUnit tests and Postman API validations, then presented the live working system at the Brunel Software Innovation Event.

Enterprise Security Assessment (Cybersecurity module)

- Applied MITRE ATT&CK threat modelling, intrusion detection and access control design to a realistic enterprise scenario aligned with CIS Controls v8.1.
- Performed quantitative risk analysis using SLE, ARO and ALE calculations to produce evidence-based security control recommendations with clear financial justification.
- Produced executive-style reporting that translated technical findings into business language for a non-specialist audience.

Applied Machine Learning Portfolio (Artificial Intelligence module, Python and R)

- Implemented clustering, classification, deep learning, neural network and Bayesian network models across a series of assessed problems.
- Applied feature engineering and cross-validation to evaluate and compare model performance against measurable baselines.
- Documented methodology and results to a reproducible research standard, including full experimental write-up.

Part Time Employment

Head Waiter – Zuma, Knightsbridge

October 2018 – Present

- Supervise and coordinate teams of 25–30 staff per shift at one of London's leading fine dining restaurants, balancing operational priorities under live service pressure.
- Trained 100+ staff members in service protocols, food safety and upselling strategies, producing structured training documentation reviewed by senior management.
- Led launch training at Zuma Saint-Tropez, France, supporting a successful international opening and developing stakeholder communication and conflict resolution skills.
- Balanced full-time study with shift work across seven years with the same employer.

Interests

Machine Learning

- Train and fine-tune transformer models outside of coursework, extending the post-OCR correction work from my dissertation onto new datasets and model architectures.
- Build small end-to-end projects using retrieval-augmented generation and prompt engineering to compare how different approaches perform on my own data.